
QS 501P : Experiments in Quantum Optics

Course Code : QS 501P

Course Name : Experiments in Quantum Optics

L-T-P-C : 0-0-5-3

Intended for : B.Tech, M.Tech, M.Sc, Ph.D.

Prerequisite : First course on Quantum Mechanics

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- NA

Laboratory/practical/tutorial Modules:

1. Fourier Transform: To demonstrate the Fourier Transform of sample images
2. Optical Tweezers: Demonstrate microscale particle manipulation dynamics through laser induced forces using Portable Optical Tweezers.
3. Optical Vortices: Demonstrate the Optical Vortices corresponding to various topological charges.
4. Quantum Eraser: To demonstrate the sensitivity of path information on the measurement process
5. Polarized 3D Cinema: Create a 3D Projection using circularly polarized light based on Real 3D Cinema Technology
6. Bomb Tester: Experiment on Interaction-free Quantum measurements: Testing the Bombs in Michelson Interferometer
7. Quantum Cryptography: Demonstration of BB84 Quantum Cryptography Protocol and detection of eavesdropper on Encrypted data

Textbooks:

1. Christopher Garry, **Introductory Quantum Optics**, Cambridge University Press, 2012.
2. Mark Fox, **Quantum Optics: An Introduction, Oxford master series in Physics**, 2006.
3. Laboratory manual for experiments

References:

1. NA